How Thawing and Cooking Interact to Affect Chicken Breast Juiciness

Final Report

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Experimental Design and Analysis of Variance

June 19, 2026

I. Research Overview

For college students managing their diet and health by eating chicken breast daily, the question of 'how can we cook chicken breast to keep it juicy and delicious?' is a very practical problem. This study applies the rigorous scientific methodology of Design of Experiments (DOE) to statistically resolve this everyday problem.

Specifically, we investigate how two common household preparation factors — Thawing Method (Factor A) and Cooking Method (Factor B) — affect the moisture retention of chicken breast, and whether there is an interaction effect between the two factors.

Rationale for Response Variable Selection

The primary response variable of this study is Cooking Loss (%). Cooking loss is an objective indicator reflecting the degree of moisture and dissolved components lost during cooking, and has been reported to be closely related to the juiciness of chicken breast. In particular, sensory moisture release and mouthfeel of chicken breast show a high correlation with cooking loss, moisture content, and water-holding capacity, making it valid to use cooking loss as a surrogate indicator for juiciness. Accordingly, this study uses cooking loss as the primary response variable and sensory evaluation as a secondary response variable, verifying the consistency between the two indicators.

II. Experimental Design

2-1. Overview of Experimental Structure

Item	Details
Research Question	How do thawing method and cooking method interact to affect the moisture retention (juiciness) of chicken breast?
Factor A (Thawing)	Refrigerator thawing (24 h, target 2–4°C) / Rapid microwave thawing (target 2–4°C)
Factor B (Cooking)	Boiling / Microwaving / Pan-frying (stopping condition: core temperature 75–80°C)
Design Structure	2×3 Factorial RCBD — Block = Day (4 days), Total N = 24 runs
Experimental Unit	CJ Cheil Jedang chicken breast cube: 4cm × 4cm × 3cm, 30g
Primary Response	Cooking Loss (%) = (Weight Before − Weight After) / Weight Before × 100
Secondary Response	Sensory evaluation: Juiciness, Hardness, Textural Preference, Overall Liking (7-point scale)
Randomization	Daily lot-drawing to determine run order of 6 treatment combinations; fully randomized within each block (day)

2-2. Standardization of Experimental Units

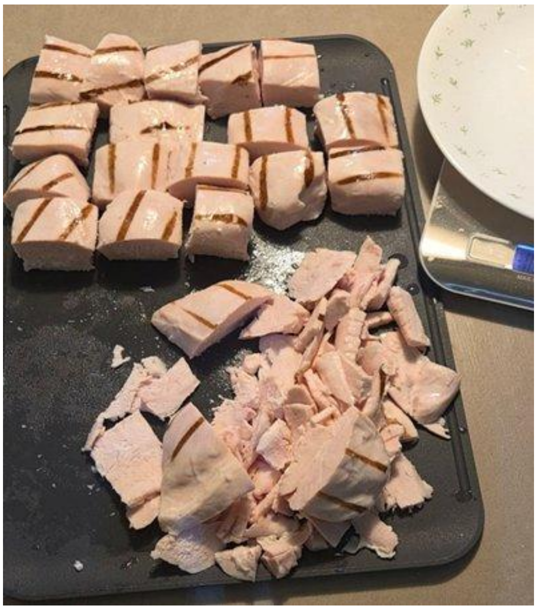
To eliminate variance from differences in thickness and surface area, all frozen chicken breasts were trimmed into geometrically identical cubes of 4cm(W)×4cm(L)×3cm(H), weighing 30g.

To ensure uniform size, frozen chicken breasts were trimmed immediately after purchase and then frozen.
(Cutting after freezing is physically difficult and weight verification becomes impractical.)

- Trimming tools: Ruler, kitchen knife
- Tolerance: ±2mm per side; weight range 30 ± 0.5g
- Measurement: Digital kitchen scale accurate to 0.01g

- Cubes outside tolerance were discarded and re-trimmed from new pieces

[Figure 1] Chicken breast cubes trimmed to uniform size and weight — sealed before freezing



[Figure 2] Trimming process of chicken breast cubes

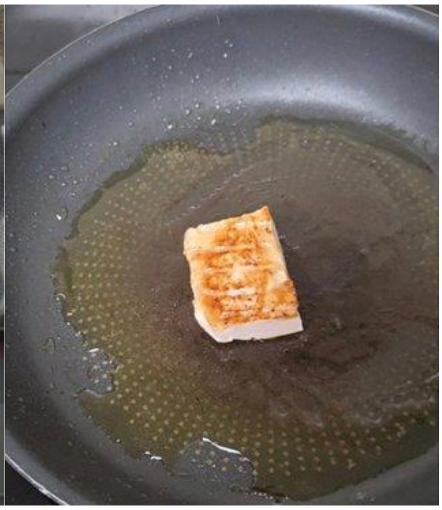
2-3. Standard Cooking Protocols

The stopping condition for all cooking methods is reaching a core temperature of 75–80°C (verified with a probe thermometer).

- [Boiling] Induction level 8; chicken added once water reaches a rolling boil; flipped once front and back every 30 seconds
- [Pan-frying] 3mL olive oil; induction level 8; flipped every 30 seconds
- [Microwaving] 700W; 30 s front side → flip → 30 s back side (1 min total; confirmed by pilot test)
- After cooking, the sample was immediately removed and rested on a room-temperature plate (22°C) for 2 minutes before weighing

[Figure 3] Boiling

[Figure 4] Microwaving



[Figure 5] Pan-frying

2-4. Sensory Evaluation Protocol

Sensory evaluation was conducted by Lee Hyunseung (single evaluator) under blinded conditions, strictly following the procedure below.

- Sample codes: Run order numbers (1–6) drawn by lot each day; cooking methods executed in that random order
- Evaluation timing: After 1-minute rest following each cooking session

- Cleansing: Rinse mouth with purified water after each sample

7-Point Scale Criteria

Item	1 point	4 points	7 points	Note
Juiciness	Extremely dry (no moisture)	Neutral	Extremely juicy (full of juice)	Higher = better
Hardness	Extremely tender	Neutral	Extremely tough (rubber-like)	Lower = better
Textural Preference	Extremely poor	Neutral	Extremely good	Higher = better
Overall Liking	Extremely dissatisfied	Neutral	Extremely satisfied	Higher = better

Sensory Evaluation Form Used in This Study

[Sensory Evaluation Form] Quality Assessment of Chicken Breast by Thawing and Cooking Combination

• Evaluation Date: 2026 / ___ / ___ (Day ___)

• Evaluator: Lee Hyunseung

• Cleansing Protocol: Before tasting each sample, thoroughly rinse mouth with the provided purified water and rest for at least 1 minute before proceeding to the next sample.

※ After confirming the sample number, please check (✓) one box per item.

1. Juiciness — Degree of juice and moisture sensation in the mouth

[1] Extremely dry (no moisture) [2] Dry [3] Slightly dry [4] Neutral [5] Slightly juicy [6] Juicy [7] Extremely juicy (full of juice)

2. Hardness — Degree of resistance when biting or chewing

[1] Extremely tender (no resistance) [2] Tender [3] Slightly tender [4] Neutral [5] Slightly tough [6] Tough [7] Extremely tough (rubber-like)

3. Textural Preference — Overall structural satisfaction considering both juiciness and hardness

[1] Extremely poor [2] Poor [3] Slightly poor [4] Neutral [5] Slightly good [6] Good [7] Extremely good

4. Overall Liking — Overall palatability considering taste, aroma, and texture

[1] Extremely dissatisfied [2] Dissatisfied [3] Slightly dissatisfied [4] Neutral [5] Slightly satisfied [6] Satisfied [7] Extremely satisfied (commercially viable)

2-5. Actual Run Order and Experimental Results (RCBD)

A RCBD with Day as the blocking factor was applied.

The table below reflects the actual run order determined by daily lot-drawing, along with the experimental and sensory evaluation results.

Block (Day)	Run Order	Thawing Method	Cooking Method	Cook Time	Before (g)	After (g)	Loss (%)	Juic	Hard	Pref	Like
Day 1	1	Refrigerator	Pan-frying	7m 20s	30g	24g	20.00%	5	3	6	6
Day 1	2	Microwave	Microwaving	1m 00s	30g	23g	23.33%	3	5	3	4
Day 1	3	Refrigerator	Microwaving	1m 00s	30g	22g	26.67%	2	6	4	3
Day 1	4	Microwave	Boiling	7m 19s	30g	24g	20.00%	5	3	4	5

Day 1	5	Refrigerator	Boiling	7m 25s	30g	24g	20.00%	3	2	4	4
Day 1	6	Microwave	Pan-frying	8m 20s	30g	26g	13.33%	7	4	7	7
Day 2	1	Microwave	Microwaving	1m 00s	30g	24g	20.00%	3	6	3	4
Day 2	2	Refrigerator	Boiling	8m 15s	30g	23g	23.33%	4	3	5	4
Day 2	3	Microwave	Pan-frying	9m 10s	30g	25g	16.67%	6	4	7	6
Day 2	4	Refrigerator	Microwaving	1m 00s	30g	22g	26.67%	2	6	3	3
Day 2	5	Microwave	Boiling	7m 16s	30g	26g	13.33%	5	4	4	5
Day 2	6	Refrigerator	Pan-frying	7m 34s	30g	25g	16.67%	5	3	6	6
Day 3	1	Refrigerator	Microwaving	1m 00s	30g	22g	26.67%	2	5	4	3
Day 3	2	Microwave	Boiling	7m 10s	30g	25g	16.67%	4	3	5	5
Day 3	3	Refrigerator	Pan-frying	8m 00s	30g	26g	13.33%	6	4	6	7
Day 3	4	Microwave	Microwaving	1m 00s	30g	26g	13.33%	4	5	4	4
Day 3	5	Refrigerator	Boiling	7m 10s	30g	26g	13.33%	3	2	4	4
Day 3	6	Microwave	Pan-frying	8m 00s	30g	26g	13.33%	7	3	7	7
Day 4	1	Microwave	Pan-frying	8m 35s	30g	26g	13.33%	7	4	6	7
Day 4	2	Refrigerator	Pan-frying	7m 50s	30g	25g	16.67%	5	3	5	6
Day 4	3	Refrigerator	Boiling	7m 40s	30g	24g	20.00%	3	3	4	3
Day 4	4	Microwave	Microwaving	1m 00s	30g	24g	20.00%	3	5	3	3
Day 4	5	Refrigerator	Microwaving	1m 00s	30g	21g	30.00%	1	7	3	2
Day 4	6	Microwave	Boiling	7m 25s	30g	25g	16.67%	6	3	4	5

2-6. ANOVA Degrees of Freedom Partition (N=24, 4 Days)

Source of Variation	df	Formula
Block (Day)	3	(4−1)
Thawing — Main Effect (A)	1	(2−1)
Cooking — Main Effect (B)	2	(3−1)
Interaction (A×B)	2	(2−1)(3−1)
Error (Residuals)	15	(4−1)(6−1)
Total	23	

2-7. Prior Justification for Day-level Blocking

Three reasons were identified for expecting the block effect to be significant:

(1) Chicken breast batch variation

Although all chicken breasts used in the experiment were purchased on the same day, there is a possibility that pieces from different batches — with varying manufacture dates, freezing durations, and storage histories — are mixed together. These differences in the chicken breast are a potential nuisance factor that could cause systematic day-to-day variation in cooking loss.

(2) Ambient temperature and humidity variation

The kitchen temperature and humidity differ slightly from day to day, which can affect the rate of heat transfer and moisture evaporation under pan-frying and boiling conditions. In particular, pan-frying is sensitive to ambient humidity and can produce day-to-day differences in cooking loss even under identical treatment conditions.

(3) Experimenter physiological state

As a single experimenter (Lee Hyunseung) is responsible for all cooking and sensory evaluation, day-to-day fluctuations in sensory sensitivity (taste baseline) and cooking proficiency may cause between-day variation in sensory scores. Note that this source of variability is expected to affect the secondary response variable (sensory evaluation) more directly than the primary response (cooking loss), since cooking loss is an objective physical measurement. Because this variation is difficult to control in advance and represents a systematic fluctuation of moderate magnitude, blocking by day is the appropriate strategy.

Post-hoc Validation ANOVA results showed the block (Day) effect was significant at $p=0.0481$, confirming the three a priori justifications with observed data. RE=1.3043 (Section X) demonstrates that the RCBD was approximately 30.4% more efficient than a CRD would have been.

2-8. Nuisance Control and the Role of Randomization

- Single experimenter: Lee Hyunseung conducted all cooking and sensory evaluation over 4 days alone
- Standardized equipment: One refrigerator, one microwave, one frying pan, one induction burner — all fixed throughout
- Palate-cleansing protocol (active elimination of evaluator fatigue): Rinse mouth with purified water + 1-minute timed rest after each sample
- Cookware heat dissipation protocol (cookware analogous measure): To prevent residual heat accumulation from consecutive use, after pan-frying the pan was immediately washed with cold water and dried to restore it to room temperature (22°C) before the next run. After boiling, the water was replaced and re-boiled. After microwaving, the door was left open for approximately 1 minute to ventilate the interior. The initial temperature of each appliance was confirmed with a digital thermometer and recorded.

Note: Randomization does NOT 'eliminate' residual heat accumulation or evaluator fatigue. What randomization provides is protection against confounding: it converts those nuisance sources from systematic bias into random error, thereby protecting the validity of inferences. Active elimination requires separate protocols — the cleansing protocol targets evaluator fatigue; the cookware cooling protocol targets residual heat.

2-9. Experimental Environment Log

Item	Day 1 (6/12)	Day 2 (6/13)	Day 3 (6/14)	Day 4 (6/16)
Start Time	10:00 AM	10:15 AM	10:05 AM	09:55 AM
End Time	12:15 PM	12:35 PM	12:10 PM	12:15 PM
Room Temp (°C)	22.1	22.0	22.3	22.0
Protocol Deviations	None	None	None	None

III. Formal Statistical Model and Analysis Plan

3-1. Mathematical Model (RCBD)

The formal statistical model reflecting the RCBD structure of this experiment is:

$$Y_{ijl} = \mu + \rho_l + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijl}$$

Symbol	Definition
Y_{ijl}	Cooking loss (%) in the l-th block (day), i-th thawing method, j-th cooking method

μ	Overall mean cooking loss
ρl	l-th block (day) effect (l = 1, 2, 3, 4)
τi	Main effect of Factor A (Thawing Method) (i = 1: Refrigerator thawing, 2: Microwave thawing)
βj	Main effect of Factor B (Cooking Method) (j = 1: Boiling, 2: Microwaving, 3: Pan-frying)
(τβ)ij	Interaction effect of the two factors — the primary hypothesis of this study
εijl	Error term: $\varepsilon_{ijl} \sim$ i.i.d. $N(0, \sigma^2)$

Constraints: $\Sigma_i\tau_i = 0$, $\Sigma_j\beta_j = 0$, $\Sigma_i(\tau\beta)_{ij} = 0$ for all j, $\Sigma_j(\tau\beta)_{ij} = 0$ for all i, $\Sigma_l\rho_l = 0$

3-2. Hypotheses

Hypothesis	Null (H_0)	Alternative (H_1)
Interaction (primary)	$(\tau\beta)_{ij} = 0$ for all i, j	At least one $(\tau\beta)_{ij} \neq 0$
Thawing Main Effect (A)	$\tau_i = 0$ for all i	At least one $\tau_i \neq 0$
Cooking Main Effect (B)	$\beta_j = 0$ for all j	At least one $\beta_j \neq 0$

3-3. Software and Packages

All statistical analyses were conducted using R.

Package	Purpose	Key Function
base R	RCBD two-way ANOVA model fitting	aov(loss ~ thaw * cook + day, data=df)
car	Levene's test for variance homogeneity	leveneTest(loss ~ thaw:cook, data=df)
emmeans	Estimated marginal means and Tukey post-hoc	emmeans(model, ~ thaw * cook)
stats	Shapiro-Wilk normality test	shapiro.test(residuals(model))

IV. Power Analysis

4-1. Parameter Derivation

Based on a pilot experiment (n=6 cubes) and Jeon et al. (2014) (SD ≈ 2–4%), σ = 3.5% was adopted as the prior estimate.

Parameter	Derivation	Value
σ (pooled SD)	Pilot experiment based; within range reported in literature	3.5%
MDE (min. detectable effect)	Cooking loss difference corresponding to 1–2 point shift on juiciness scale	5.0%p
Cohen's d	MDE / σ = 5.0 / 3.5	1.4286
Cohen's f	d / 2	0.7143
Cohen's f^2	f^2	0.5102
λ (noncentrality, N=24)	$f^2 \times N = 0.5102 \times 24$	12.2449

4-2. Final Power Results (N=24, 4 Days)

Effect	df_num	df_error	λ(NCP)	Power
Thawing Main Effect (A)	1	15	12.2449	0.9046
Cooking Main Effect (B)	2	15	12.2449	0.8136
Interaction (A×B)	2	15	12.2449	0.8136

With N=24 (4-day blocks), the primary hypothesis — the interaction effect — achieves a power of 0.8136, meeting the conventional threshold of 0.80.

V. Statistical Assumption Testing

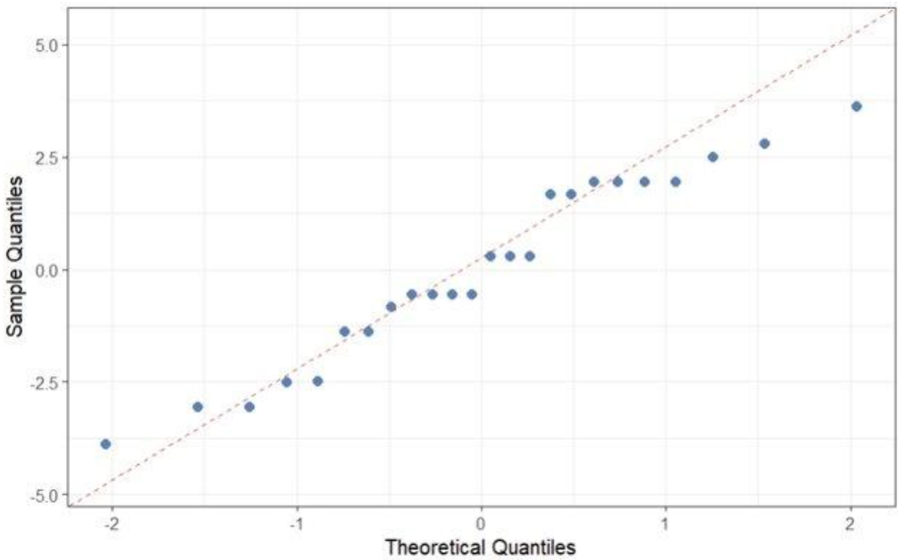
Test	Response Variable	Statistic	p-value
Shapiro-Wilk (normality)	Cooking loss residuals	W = 0.9552	0.3497
Shapiro-Wilk (normality)	Juiciness residuals	W = 0.9615	0.4690
Levene's Test (homogeneity)	Cooking loss	F = 0.3995	0.8428

Both cooking loss residuals (W=0.9552, p=0.3497) and juiciness residuals (W=0.9615, p=0.4690) satisfy the normality assumption, and homogeneity of variance is also satisfied (p=0.8428). Analysis proceeds on the original scale without transformation.

5-1. Residual Diagnostics

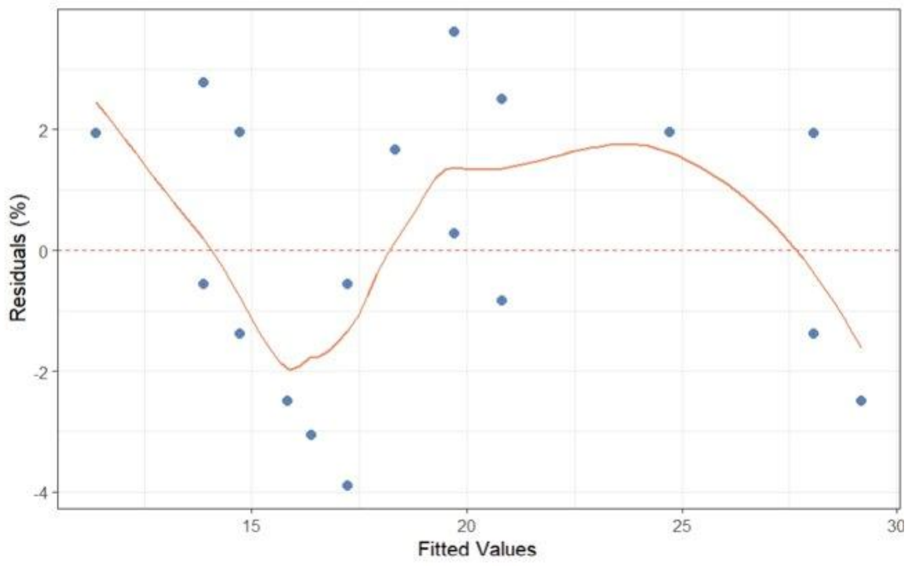
Figures 6–8 show the residual diagnostic results for the RCBD model (fit loss).

[Figure 6] Normal Q-Q Plot



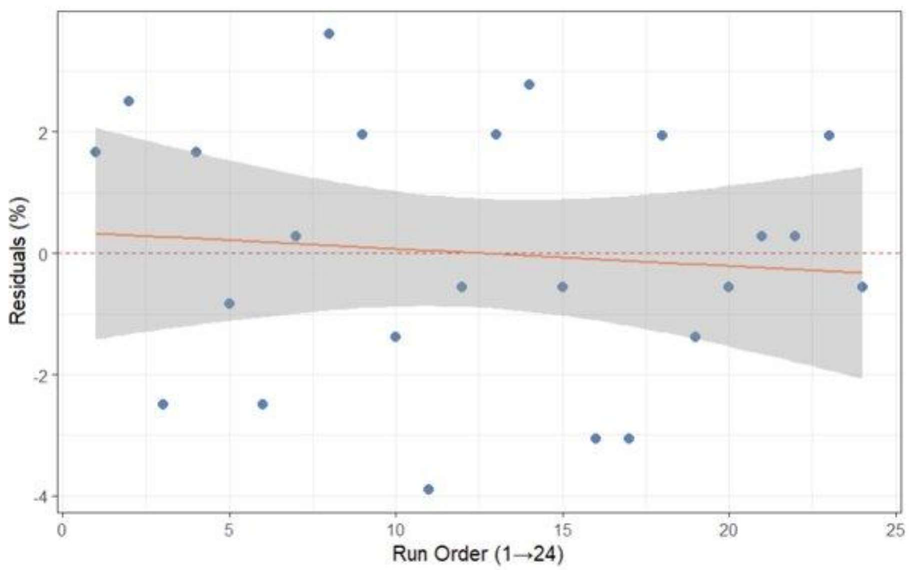
Residuals generally follow the theoretical quantiles across the full range. One extreme value on the lower left shows minor deviation, but Shapiro-Wilk (W=0.9552, p=0.3497) confirms normality is satisfied.

[Figure 7] Residuals vs Fitted



The LOESS curve shows slight nonlinearity but no pronounced heteroscedastic pattern. Levene's Test (p=0.8428) formally confirms the homogeneity of variance assumption.

[Figure 8] Residuals vs Run Order



The regression line is nearly horizontal (slope ≈ 0) and the 95% confidence interval contains zero throughout. No evidence of temporal drift; independence of observations is supported.

VI. Descriptive Statistics

Thawing	Cooking	n	Loss Mean (%)	Loss SD	Juiciness	Hardness	Overall Liking
Refrigerator	Boiling	4	19.17	4.19	3.25	2.50	3.75
Refrigerator	Microwaving	4	27.50	1.66	1.75	6.00	2.75
Refrigerator	Pan-frying	4	16.67	2.72	5.25	3.25	6.25
Microwave	Boiling	4	16.67	2.72	5.00	3.25	5.00
Microwave	Microwaving	4	19.17	4.19	3.25	5.25	3.75
Microwave	Pan-frying	4	14.17	1.67	6.75	3.75	6.75

The worst combination — Refrigerator × Microwaving — recorded the highest cooking loss (27.50%) and the lowest juiciness score (1.75/7). Both response variables consistently point to this combination as the most detrimental to moisture retention.

The optimal combination — Microwave × Pan-frying — recorded the lowest cooking loss (14.17%) and the highest juiciness score (6.75/7). The perfect inverse correspondence between the two measures confirms this as the optimal preparation for juicy chicken breast.

The gap between the two extremes is 13.33%p in cooking loss and 5.0 points in juiciness, demonstrating that the choice of preparation method has a practically large impact on eating quality.

VII. Two-Way ANOVA Results (RCBD)

7-1. Primary Response: Cooking Loss (%)

Source	df	SS	MS	F	p-value
Block (Day)	3	66.70	22.23	3.333	0.0481
Thawing — Main Effect (A)	1	118.59	118.59	17.780	0.0007
Cooking — Main Effect (B)	2	262.10	131.05	19.647	< 0.0001
Interaction (A×B)	2	45.44	22.72	3.406	0.0603 .
Error (Residuals)	15	100.05	6.67		
Total	23	592.88			

RCBD model results showed the block (Day) effect was significant ($p=0.0481$), confirming systematic between-day variability and validating the RCBD design choice post-hoc. Thawing method (A) was highly significant ($p=0.0007$): microwave thawing produced approximately 4.2%p lower cooking loss than refrigerator thawing, demonstrating that thawing method independently contributes to moisture retention. Cooking method (B) was the stronger factor ($p<0.0001$): the gap between microwaving (23.33%) and pan-frying (15.63%) was 7.70%p, confirming cooking method as the dominant determinant of moisture loss in this experiment. The interaction (A×B) was not significant at $\alpha=0.05$ ($p=0.0603$).

7-2. Interaction Interpretation

The interaction $p=0.0603$ did not reach statistical significance at $\alpha=0.05$. However, from an effect size perspective, $\eta^2 = 45.44/592.88 = 0.077$ indicates the interaction explains approximately 7.7% of total variance — a level that is difficult to dismiss as practically negligible. In particular, in the microwaving condition, the cooking loss gap between refrigerator thawing (27.50%) and microwave thawing (19.17%) was 8.33%p, markedly larger than in the boiling (2.50%p) and pan-frying (2.50%p) conditions, suggesting that statistical significance could be confirmed with a larger sample ($n\geq6$ per cell).

7-3. Secondary Response: Juiciness

Source	df	SS	MS	F	p-value
Block (Day)	3	0.13	0.042	0.111	0.9520
Thawing (A)	1	15.04	15.042	40.111	< 0.0001
Cooking (B)	2	49.08	24.542	65.444	< 0.0001

Interaction (A×B)	2	0.08	0.042	0.111	0.8956
Error (Residuals)	15	5.63	0.375		

Juiciness ANOVA results showed highly significant main effects for both thawing (A, p<0.0001) and cooking method (B, p<0.0001). The block effect (p=0.9520) and interaction (p=0.8956) were both non-significant, showing that the two factors act independently of one another regardless of day. The fact that the juiciness interaction is fully non-significant (p=0.8956) while the cooking-loss interaction is borderline (p=0.0603) shows that the conclusions from the two response variables are broadly consistent, with only a minor discrepancy.

7-4. Secondary Responses: Hardness and Overall Liking

Hardness

Source	df	SS	MS	F	p-value
Block (Day)	3	1.67	0.556	1.923	0.169
Thawing (A)	1	0.17	0.167	0.577	0.459
Cooking (B)	2	33.25	16.625	57.548	< 0.0001
Interaction (A×B)	2	2.58	1.292	4.471	0.030
Error (Residuals)	15	4.33	0.289		

For hardness, cooking method (B) showed overwhelming influence (F=57.55, p<0.0001), with microwaving producing the highest toughness. Notably, the interaction was significant (p=0.030): refrigerator thawing combined with microwaving produced a pronounced increase in toughness. This means the interaction that was borderline for cooking loss and juiciness is statistically clear in the hardness dimension, further confirming that the refrigerator thawing + microwaving combination is the most disadvantageous not only for moisture loss but also for textural firmness.

Overall Liking

Source	df	SS	MS	F	p-value
Block (Day)	3	1.46	0.486	3.182	0.0546
Thawing (A)	1	5.04	5.042	33.000	< 0.0001
Cooking (B)	2	43.58	21.792	142.636	< 0.0001
Interaction (A×B)	2	0.58	0.292	1.909	0.1825
Error (Residuals)	15	2.29	0.153		

For overall liking, both thawing method (p<0.0001) and cooking method (p<0.0001) had significant effects, while the interaction was completely non-significant (p=0.1825), showing that the two factors act independently on consumer liking. This means overall liking is determined by the main effects of each factor independently, with no synergistic or antagonistic combination effects.

VIII. Tukey HSD Post-Hoc Comparisons

Since the interaction (cooking loss, overall liking) was not significant, pairwise comparisons were conducted for main effects and treatment combinations. Only significant pairs (p<0.05) are shown below.

Contrast	Difference (%p)	t-ratio	p (Tukey)
Ref+Micro vs Ref+Boil	+8.34	4.565	0.0041

Ref+Micro vs Mic+Boil	+10.84	5.933	0.0003
Ref+Micro vs Ref+Pan	+10.84	5.933	0.0003
Ref+Micro vs Mic+Pan	+13.34	7.303	< 0.0001
Ref+Micro vs Mic+Micro	+8.34	4.565	0.0041

Tukey HSD results showed significant differences in 5 of 15 pairwise comparisons, all involving the Refrigerator × Microwaving combination. Refrigerator × Microwaving showed significant differences from Refrigerator+Boiling (+8.34%p, p=0.0041), Microwave+Boiling (+10.84%p, p=0.0003), Refrigerator+Pan-frying (+10.84%p, p=0.0003), Microwave+Pan-frying (+13.34%p, p<0.0001), and Microwave+Microwaving (+8.34%p, p=0.0041), confirming it as the worst combination with the highest cooking loss among the 6 combinations. Microwave × Pan-frying had the lowest estimated marginal mean (EMM = 14.17%) and the highest juiciness score (6.75), confirming it as the optimal combination for moisture retention.

IX. Interaction Plots and Main Effect Plots

9-1. Interaction Plots

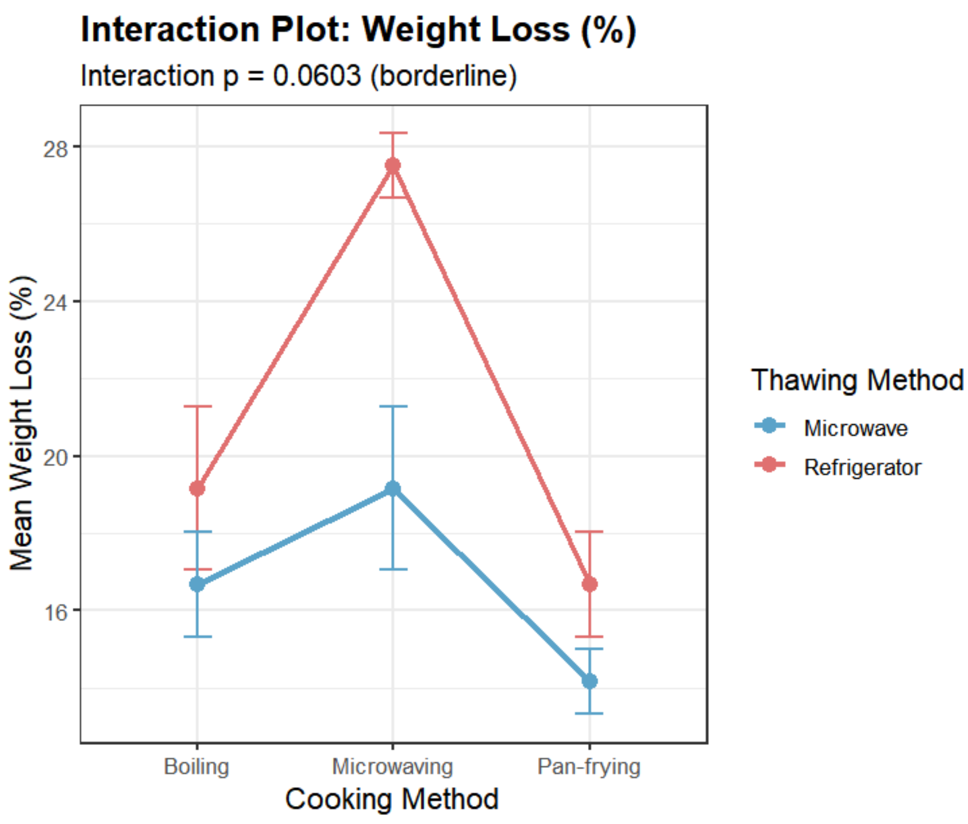


Figure 9. Interaction Plot: Cooking Loss (%) — Thawing × Cooking (p = 0.0603, borderline). The two lines are generally parallel, but in the Microwaving condition the refrigerator thawing loss rate (27.50%) is markedly higher than microwave thawing (19.17%), visually confirming the borderline interaction. Ordinal interaction pattern — no crossover observed.

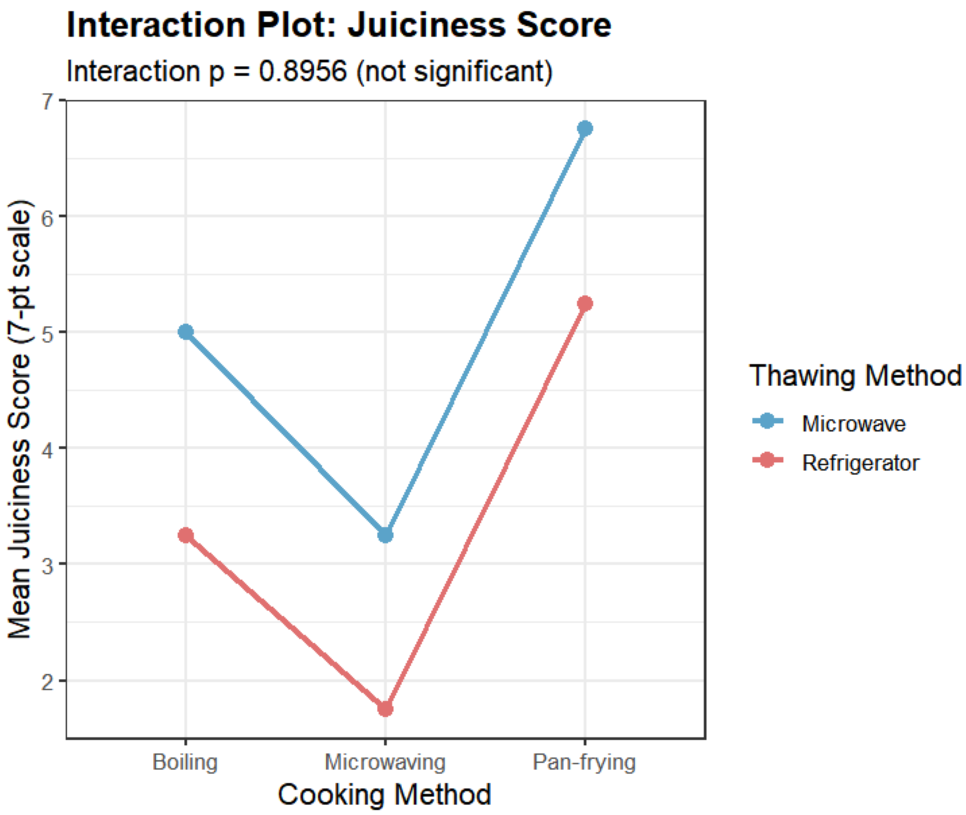


Figure 10. Interaction Plot: Juiciness Score — Thawing × Cooking (p = 0.8956, not significant). The two lines are completely parallel, clearly showing the absence of interaction for juiciness. Conclusions are consistent across both response variables.

9-2. Thawing Method — Main Effect Plots

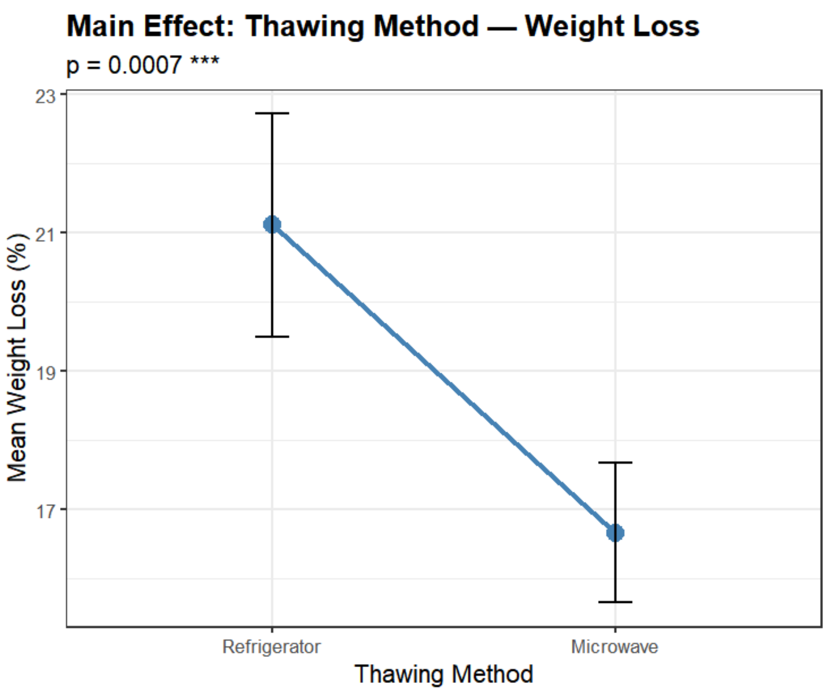


Figure 11. Main Effect: Thawing Method — Cooking Loss (p = 0.0007)

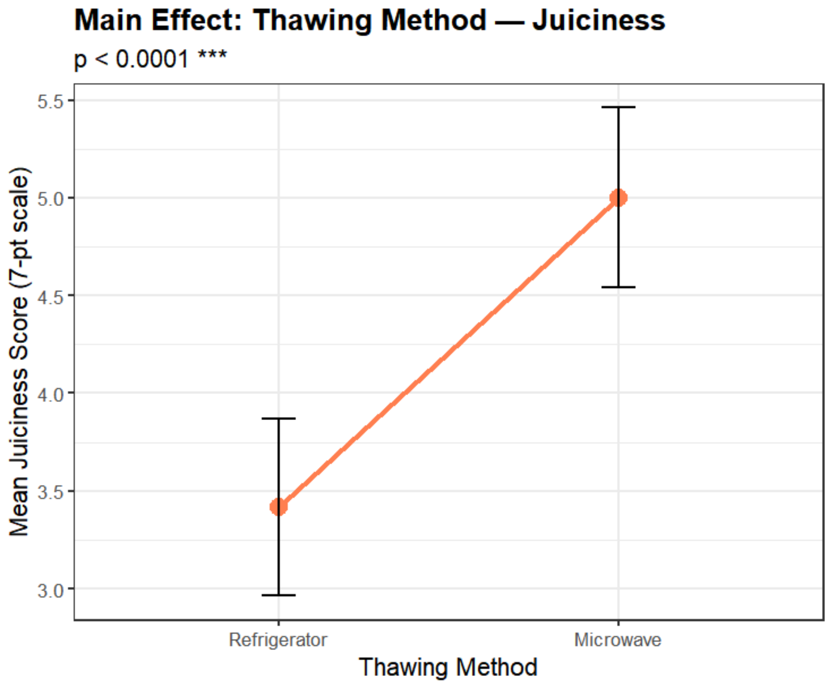


Figure 12. Main Effect: Thawing Method — Juiciness ($p < 0.0001$)

Microwave thawing (16.8%) produced approximately 4.2%p lower cooking loss than refrigerator thawing (21.0%), and juiciness score was also higher at 5.0 points compared to refrigerator thawing (3.4 points). Both response variables consistently support the superiority of microwave thawing.

9-3. Cooking Method — Main Effect Plots

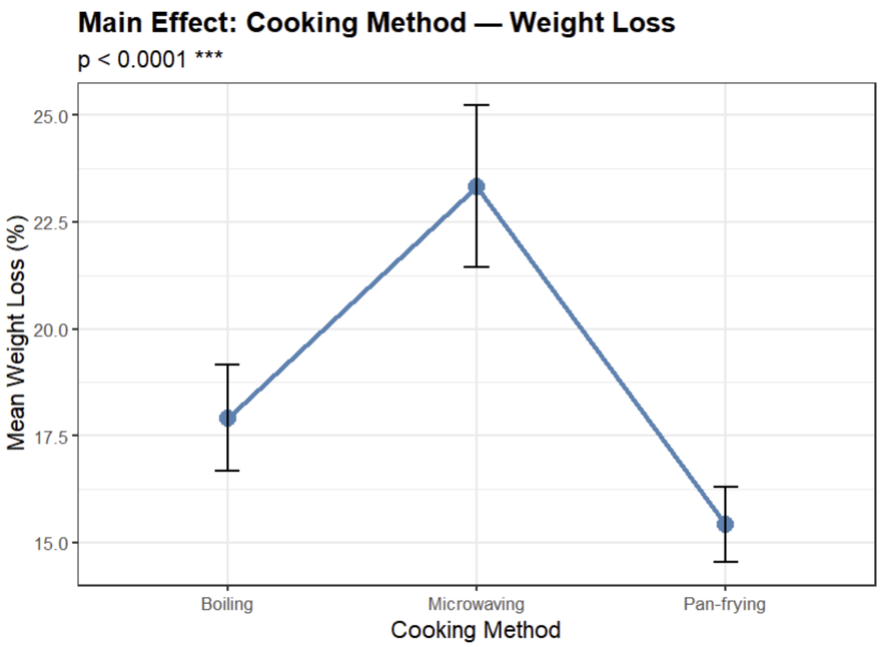


Figure 13. Main Effect: Cooking Method — Cooking Loss ($p < 0.0001$)

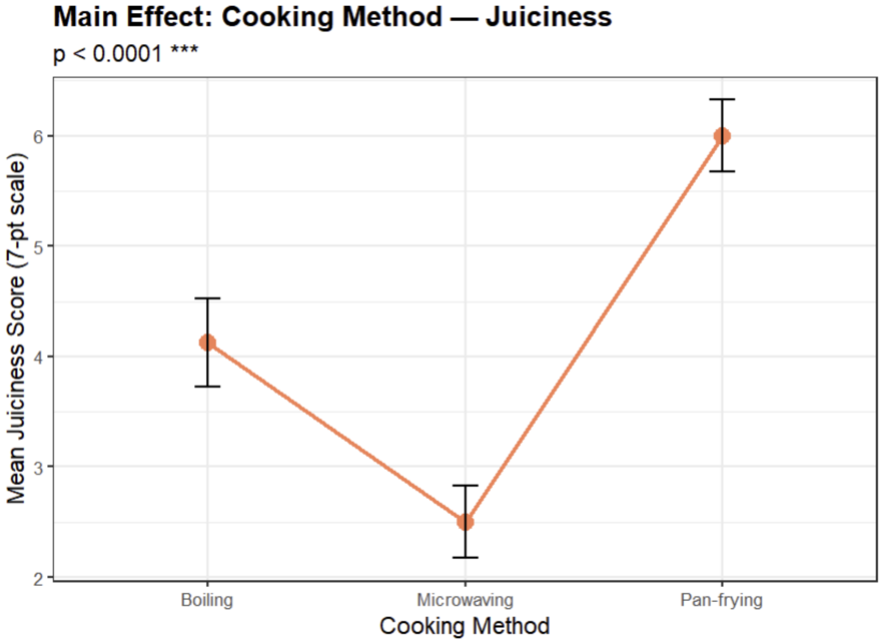


Figure 6. Main Effect: Cooking Method — Juiciness (p < 0.0001)

Cooking loss ranks Microwaving (23.3%) > Boiling (18.0%) > Pan-frying (15.6%) (Figure 5), while juiciness is perfectly inverse: Pan-frying (6.0) > Boiling (4.1) > Microwaving (2.5) (Figure 6). Cooking method is confirmed as a stronger influence factor than thawing method in both response variables (FB=19.65 vs FA=17.78).

X. Relative Efficiency of RCBD vs CRD (RE)

MS_Block	MS_Error	RE Formula	RE Value
22.23	6.67	[(r−1)·MSB + r(k−1)·MSE] / [(rk−1)·MSE]	1.3043

Using MS_Block=22.23 and MS_Error=6.67 extracted from the ANOVA results, the relative efficiency of RCBD compared to CRD was calculated.

RE = [(r−1)×MS_Block + r(k−1)×MS_Error] / [(rk−1)×MS_Error]

= [(4−1)×22.23 + 4×(6−1)×6.67] / [(4×6−1)×6.67]

= [66.69 + 133.40] / 153.41 = 1.3043

RE=1.3043 means a CRD would require approximately 30.4% more experimental units to achieve the same precision as the RCBD. In other words, this experiment's RCBD with N=24 achieved the same statistical precision as a CRD with N≈31.

This result is consistent with the ANOVA finding that the block (Day) effect was significant at p=0.0481. The significance of the block effect statistically confirms that systematic between-day variability was real, and RE>1.0 demonstrates through post-hoc data that absorbing that variability into the block term resulted in a substantial reduction in error variance. Had a CRD without blocking been applied, the day-level variability would have been included in the error term, inflating MS_Error, reducing F-statistics, and diminishing the ability to detect the main and interaction effects. In conclusion, the RE analysis provides post-hoc empirical validation of the statistical appropriateness of the RCBD design choice.

XI. Conclusions and Implications

11-1. Summary of Key Findings

Category	Conclusion
Interaction (A×B)	Not significant at $\alpha=0.05$ ($p=0.0603$). $\eta^2=0.077$ explains 7.7% of total variance. Gap in Microwaving condition (8.33%p) may become significant with larger samples.
Optimal Combination	Microwave + Pan-frying: lowest cooking loss (14.17%), highest juiciness (6.75/7)
Worst Combination	Refrigerator + Microwaving: highest cooking loss (27.50%), lowest juiciness (1.75/7)
Dominant Factor	Cooking method > Thawing method (FB=19.65 vs FA=17.78)
Hardness Interaction	Interaction significant ($p=0.030$): Refrigerator+Microwaving produces especially tough texture
RCBD Efficiency	RE=1.3043 → 30.4% more efficient than CRD. Block effect significant ($p=0.0481$) validates design.

This study investigated the effects of thawing method (refrigerator vs microwave thawing) and cooking method (boiling, microwaving, pan-frying) on moisture retention of chicken breast using a 2×3 RCBD design (N=24, 4-day blocks), and the key findings are as follows.

The primary hypothesis — the interaction effect (A×B) — did not reach statistical significance ($p=0.0603$), but an effect size of $\eta^2=0.077$ explaining 7.7% of total variance was observed, a level difficult to dismiss as practically negligible. In particular, the cooking loss gap between refrigerator and microwave thawing in the microwaving condition (8.33%p) was markedly larger than in other cooking method conditions (2.50%p), suggesting significance could be confirmed with a larger sample.

Main effect analysis confirmed that cooking method ($F=19.65$) was a stronger influence factor than thawing method ($F=17.78$), suggesting that cooking method selection is more decisive for moisture retention. In the secondary response of hardness, the interaction was clearly significant ($p=0.030$), further confirming that the refrigerator thawing + microwaving combination is the most disadvantageous not only for moisture loss but also for textural firmness. In terms of design efficiency, RE=1.3043 demonstrated approximately 30.4% higher statistical efficiency than a CRD, and the significance of the block effect ($p=0.0481$) fully supports the appropriateness of the RCBD design choice post-hoc.

11-2. Practical Recommendations

Based on the findings of this study, the following practical guidelines are offered for cooking chicken breast at home.

If moisture retention is the top priority, the microwave thawing + pan-frying combination is strongly recommended. This combination achieved optimal results in both response variables with a cooking loss of 14.17% and a juiciness score of 6.75. Conversely, microwaving as a cooking method should be avoided regardless of thawing choice, as it produces the highest cooking loss (23.33%) and dry texture. Even with refrigerator thawing, combining it with pan-frying or boiling can preserve considerable moisture, so it is worth noting that cooking method selection plays a more decisive role than thawing choice. For those seeking a tender texture, boiling recorded the lowest hardness score at an average of 2.5 points, and the cooking method can be varied according to individual texture preferences.

11-3. Limitations and Future Directions

This study has several methodological limitations.

First, sensory evaluation was conducted by a single evaluator (Lee Hyunseung), making inter-rater reliability verification impossible; the possibility that individual sensory standards acted as systematic bias throughout the scores cannot be ruled out. Second, the borderline interaction ($p=0.0603$) warrants further investigation: increasing replication from the current $n=4$ to $n=6$ or more per cell would enable a clearer determination of statistical significance. Third, because experimental units were limited to 30g standardized cubes, there are limitations in directly generalizing results to whole chicken breasts (~200g) used in actual homes, and differences in heat transfer characteristics by size and shape may affect outcomes.

Future research is recommended to secure three or more sensory panelists to improve evaluation reliability, increase replication to 6 or more to re-verify the significance of the interaction, and apply Response Surface Methodology (RSM) with cooking temperature and time as continuous factors to derive optimal cooking conditions more precisely.

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